

Double-click (or enter) to edit

```
from google.colab import files
uploaded = files.upload()
```

bfeedbbrEx-...R_IEnhP.jpg

bfeedbbrEx-5agUbGZrPQqVVotMwdd3tF8dQ6oCKBikDAF7-

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VO_GYxeucUy0yjL9UJtVZ4g8NjNjijJ1ib_8gQyl-joR_IEnhP.jpg(image/jpeg) - 428835 bytes, last modified: 7/16/2026 - 100% done

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```
# Step 1: Install libraries
!pip install opencv-python matplotlib

# Step 2: Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Step 3: Upload image
from google.colab import files
uploaded = files.upload()

# Step 4: Read image
img_path = list(uploaded.keys())[0]
img = cv2.imread(img_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
img = img / 255.0    # Normalize

# =====
# PART 1: CCD vs CMOS Simulation
# =====
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```
# CCD (Low noise)
ccd_noise = np.random.normal(0, 0.02, img.shape)
ccd_img = np.clip(img + ccd_noise, 0, 1)

# CMOS (High noise + pattern noise)
cmos_noise = np.random.normal(0, 0.05, img.shape)
pattern = np.tile(np.linspace(0, 0.1, img.shape[1]), (img.shape[0], 1))
pattern = np.expand_dims(pattern, axis=2)

cmos_img = np.clip(img + cmos_noise + pattern, 0, 1)

# Convert back
ccd_img_disp = (ccd_img * 255).astype(np.uint8)
cmos_img_disp = (cmos_img * 255).astype(np.uint8)

# =====
# PART 2: Distortion Simulation
# =====

# (a) Add noise
noise = np.random.normal(0, 25, img.shape).astype(np.uint8)
noisy = cv2.add((img*255).astype(np.uint8), noise)

# (b) Blur (defocus)
blur = cv2.GaussianBlur(noisy, (9,9), 0)

# (c) Low resolution (sampling issue)
small = cv2.resize(blur, (img.shape[1]//4, img.shape[0]//4))
distorted = cv2.resize(small, (img.shape[1], img.shape[0]))

# =====
# PART 3: Improvement
# =====

# Denoising
denoised = cv2.fastNlMeansDenoisingColored(distorted, None, 10, 10, 7, 21)
```

```
# Sharpening
kernel = np.array([[0,-1,0],[-1,5,-1],[0,-1,0]])
restored = cv2.filter2D(denoised, -1, kernel)

# =====
# DISPLAY RESULTS
# =====

plt.figure(figsize=(12,8))

plt.subplot(2,3,1)
plt.imshow((img*255).astype(np.uint8))
plt.title("Original")
plt.axis('off')

plt.subplot(2,3,2)
plt.imshow(ccd_img_disp)
plt.title("CCD Image (Low Noise)")
plt.axis('off')

plt.subplot(2,3,3)
plt.imshow(cmos_img_disp)
plt.title("CMOS Image (High Noise)")
plt.axis('off')

plt.subplot(2,3,4)
plt.imshow(distorted)
plt.title("Distorted Image")
plt.axis('off')

plt.subplot(2,3,5)
plt.imshow(restored)
plt.title("Improved Image")
plt.axis('off')

plt.tight_layout()
plt.show()
```


Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
 Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
 Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
 Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.63.0)
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 Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.2)
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 Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib)

KGE_5TKp...85GBjqb.jpg

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zAUguV8eSx8U85GBjqb.jpg(image/jpeg) - 40963 bytes, last modified: 7/22/2026 - 100% done

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Original



CCD Image (Low Noise)



CMOS Image (High Noise)

